

1. Raphael Talento (917451240)
2. Sarath Kavitha (918527303)

Relevant Files:

- allhars.zip (all har files, made one for each website)
- editedtop-1m.csv
- haranalysis.py
- haranalysisgpt.py
- seleniumcrawler.py
- seleniumcrawlergpt.py

About seleniumcrawler.py and seleniumcrawlergpt.py

ChatGPT conversations:

<https://chat.openai.com/share/8f015605-f088-4b0a-bc28-e882ca6271c6>

<https://chat.openai.com/share/49223f2c-5945-4bef-999b-99a35dae5941>

<https://chat.openai.com/share/5fb89911-0d3f-4898-a4f7-b0fc754cfcad>

Some issues that arose when doing the selenium crawling were: 1. Random webpages would stall and never proceed, even when try except statements and timeouts were introduced and 2. How many webpages we should include in a har file.

There was also an issue with accidentally capturing the content (the request payload) of each webpage. After the script ran, har files were 230 GB.

The GitHub for selenium crawling was sufficient in getting us started. However, the above issues were not directly addressed by ChatGPT, and instead were addressed based on advice from the professor and through experience running the script.

To truly isolate a potential problem website, we made a .har file for each website. In addition, to make sure the driver would have a “fresh” set of resources or a clean cache, we quit the driver and reopen it every time we wrote the a .har. Every time an error would arise, we would also quit the driver and reopen it.

However, we were successful in getting 1000 hars. If there are “bad hars”, (instances where the webpage wouldn’t load) our selenium crawler wouldn’t care as it would try to access some cookies head. Bad hars wouldn’t be able to get any cookie headers, and instead the contents would be something like ““status”: 404, “statusText”: “Not Found”” or “status”: 503, “statusText”: “Service Unavailable: Back-end server is at capacity”. The cookies header would often be empty with this. This is typically found in our smaller hars, (the 2kb and 3kb ones) which makes sense, as those can’t actually capture any meaningful headers or cookies.

In other words, the handling of bad hars or bad web pages was handled in our selenium crawler. To be careful, our code for har analysis would still be able to extract only relevant information from each har even if the har passed didn’t have anything meaningful.

ChatGPT was mainly used to query how to write certain things, or give some minor suggestions. We lastly passed our seleniumcrawler.py code and asked it to clean it up/optimize it into seleniumcrawlergpt.py.

However, we are not able to confirm how well the seleniumcrawlergpt.py was implemented. This would require us to run the script again, and also pass that through our har analysis. The problem with this is that the cookies that could be passed or extracted in the responses to these websites would definitely be a different number, and there's also a potential for different "bad urls" (some webpages may be down for whatever reason, or ones in our initial har extraction could be good again)

For this reason, our seleniumcrawler.py is the main and more reliable code when compared to seleniumcrawlergpt.py.

haranalysis.py results (haranalysisgpt.py results will be more accurate, as it incorporates tldextract)

Number of third-party domain requests across all sites: 60657

10 most common third-parties:

- optimizationguide-pa.googleapis.com: 1248
- cm.g.doubleclick.net: 1007
- www.googletagmanager.com: 889
- www.google-analytics.com: 851
- pixel.rubiconproject.com: 851
- cdn.cookie law.org: 823
- s.amazon-adsystem.com: 783
- fonts.gstatic.com: 713
- simage2.pubmatic.com: 673
- www.google.com: 625

Number of third party cookies: 118407

- audit: 3056
- khaos: 2785
- KADUSERCOOKIE: 2374
- chkChromeAb67Sec: 2324
- PugT: 2314
- pi: 2156
- SyncRTB3: 2086
- DPSync3: 1857
- IDE: 1583
- KRTBCOOKIE_57: 1359

Disclaimer: From personal experience, very hard to actually group these third party domains and all their variations together (versions of URLs/redirects areyoutu.be with the Youtube domain, amzn with the Amazon domain, o365 with Office 365 which is actually the Microsoft domain. As a result, every redirect to a unique url is treated differently and not counted together. But haranalysisgpt.py fixes this by importing tldextract.

The numbers of cookies seem reasonable across the 1000 websites. This is mostly because checking Cookiepedia, websites alone like google.com have 22,075 3P cookies, or Yahoo having 1401 3P cookies.

Unfortunately, most of the cookies I analyzed were revealed to be unknown cookies. It's likely that the ones applicable to this are what cookiepedia deems as "unknown cookies", where their purpose is unknown.

However, there's a section that allows us to click on a URL to get more details about a specific cookie on that specific website URL. It does provide more insight about what the cookie is used for, but the use of the cookie may differ between websites. This is probably why cookiepedia is deeming these to be unknown cookies.

I'll also be referencing other websites in case Cookipedia doesn't provide any information on the cookie gathered.

Intended Functionality of Cookies:

- audit

Main purpose of this cookie is unknown. However, (when clicking on specific websites) it is a cookie used to deliver adverts (advertisements) more relevant to the user's interest. It's also used to measure the effectiveness of an advertising campaign, as it limits the number of times you see an ad. In other words, Cookipedia hints that it's most likely related to a cookie for targeting/advertising. According to [cookiestudy.com](https://cookiestudy.com/en/cookie/audit), this is a cookie owned by Rubicon Project (Magnite). Sites online suggest that Rubicon Project that helps publishers monetize ad space, which suggests that Rubicon Project helps publishers display relevant ads to users (by tracking their interests)

- khaos

No information on this cookie for Cookipedia. However, according to [Rijkmuseum](https://rijkmuseum.nl/en/cookie/khaos), khaos is a third party cookie that registers IP address, geographic location, visited websites, and based on which ads are clicked by user, it can optimize the display of ads based on how the user traverses websites in the same ad network. According to [Accordion](https://www.cookieconsent.com/en/cookie/khaos), it's used for analytics.

- KADUSERCOOKIE

This cookie is used for targeting/advertising according to Cookipedia. Also according to [Cookiestudy.org](https://cookiestudy.com/en/cookie/kadusercookie), this is a cookie set by PubMatic. It registers a unique ID to a user and "identifies a returning user's device across websites that use the same ad network", in which the ID is used for targeted ads.

- chkChromeAb67Sec

There was virtually no information on this in Cookipedia or Cookiestudy.org. However, the cookie seems to be set by [PubMatic](https://pubm.com/en/cookie/chkchromeab67sec), as in a document written by them it says that this cookie is "set on Google Chrome browsers that have a version above 67 and is used for testing purposes".

- PugT

According to Cookiepedia, this is a cookie used for targeting/advertising. It's also generally set/provided by PubMatic. According to [PubMatic](https://pubm.com/en/cookie/pugt) as well, the cookie "tracks when cookies were

updated on the browser, in order to limit the number of calls to the server-side cookie store”. From [Intalytics](#) as well, is the exact same description.

- pi

According to Cookiepedia, the general purpose of this cookie is unknown. Clicking on sites provided by Cookiepedia, they suggest that this cookie is most likely used for targeting/advertising. Cookiesearch.org also doesn’t show much on this cookie, but sites (without explicitly saying) seem to relate this to Bombora and <https://ml314.com/>, which is a company that collects information/data on users. It supposedly collects this information for B2B (business to business) websites.

- SyncRTB3

According to Cookiepedia, the general use of this cookie is unknown. However, according to [Cookiesearch.org](#), it is a cookie used for advertisements set by PubMatic. According to [Pubmatic](#) themselves, this cookie “maintain a list of our partner’s pixel IDs, which may be used to read and place cookies on a web browser, that have been synced with PubMatic IDs. This enables us to identify the partners with whom we have already synced our cookies and to ensure that we do not unnecessarily sync cookies again within a certain period of time.”

- DPSync3

This cookie is similar to the cookie above (SyncRTB3). The exact description applies to this one as well.

- IDE

According to Cookiepedia, this cookie is used for targeting/advertising and is owned by Doubleclick (Google). It’s a real time bidding advertising exchange. According to [Cookiedatabase.org](#), this cookie is used for ad delivery or retargeting.

- KRTBCOOKIE_57

There was no information on this cookie in Cookiepedia other Cookiedatabase.org. However, according to [New Mexico Tech](#), this is a cookie by PubMatic and is for marketing and advertising purposes.

haranalysis.py output:

```
Top 15 3P Cookie Counts of the 1000 HAR files:
audit: 3056
khaos: 2785
KADUSERCOOKIE: 2374
chkChromeAb67Sec: 2324
PugT: 2314
pi: 2156
SyncRTB3: 2086
DPSync3: 1857
IDE: 1583
KRTBCOOKIE_57: 1359
A3: 1330
KRTBCOOKIE_153: 1292
KRTBCOOKIE_377: 1285
i: 1216
ad-id: 1194
```

```
Total Count of 3P Cookies: 118407
```

```
-----

Top 15 Third Parties of the 1000 HAR files:
optimizationguide-pa.googleapis.com: 1248
cm.g.doubleclick.net: 1007
www.googletagmanager.com: 889
www.google-analytics.com: 851
pixel.rubiconproject.com: 851
cdn.cookie law.org: 823
s.amazon-adsystem.com: 783
fonts.gstatic.com: 713
simage2.pubmatic.com: 673
www.google.com: 625
www.gstatic.com: 591
match.adsrvr.org: 561
securepubads.g.doubleclick.net: 555
content-autofill.googleapis.com: 530
ib.adnxs.com: 513
```

```
Total Count of third party domains: 60657
```

Disclaimer: From personal experience, very hard to actually group these third party domains and all their variations together (versions of URLs/redirects areyoutu.be with the Youtube domain, amzn with the Amazon domain, o365 with Office 365 which is actually the Microsoft domain). As a result, every redirect to a unique url is treated differently and not counted together. But haranalysisgpt.py fixes this by importing tldextract.

haranalysisgpt.py results (more accurate than haranalysis.gpt, as this version correctly counts the 3P domains)

10 most common third-parties:

- doubleclick: 2730
- googleapis: 2378
- pubmatic: 2025
- rubiconproject: 1903
- gstatic: 1389
- google: 1245
- amazon-adsystem: 1184
- yahoo: 960
- googlesyndication: 926
- googletagmanager: 891

Number of third party cookies: 118407

- audit: 3056
- khaos: 2785
- KADUSERCOOKIE: 2374
- chkChromeAb67Sec: 2324
- PugT: 2314
- pi: 2156
- SyncRTB3: 2086
- DPSync3: 1857
- IDE: 1584
- KRTBCOOKIE_57: 1359

Note: Amount of cookies are slightly higher than haranalysis.gpt. Makes sense, as it's now counting some cookies more often than haranalysis.py did.. Also note the difference of third parties being counted between haranalysisgpt.py and haranalysis.py. Haranalysisgpt.py imports [tld](#), so haranalysisgpt.py is more accurate.

ChatGPT conversations for har analysis:

<https://chat.openai.com/c/e22702dc-ee99-4043-a43b-732fbba59d60> (conversation to help me more accurately/thoroughly extract information from .har files)

<https://chat.openai.com/share/80e23453-9ae0-4862-913e-8cfaf785cd19> (specific to haranalysisgpt.py)

The goal of haranalysisgpt.py was to properly count the third-party domains. The issue is that there was a lot of variety/minor variations in URLs it redirected to, so the initial thinking of

splitting on a “.” to get some domain was not appropriate for the volume our analysis was doing. As a result, our haranalysis.py inaccurately counts the 3P domains. We really just fed our code in, asked how to use this library given our code, and made some minor changes to our code to accurately get the 3P domains.

ChatGPT was also used to figure out how to more efficiently gather the necessary/relevant information from the .har files initially. This was mainly because the structure of some .har files were massive, and debugging/confirming that what we extracted was indeed correct would be too meticulous. It was used to speed up our har analysis as a whole.

haranalysisgpt.py output:

```
Top 15 Cookie Counts of the 1000 HAR files:
audit: 3056
khaos: 2785
KADUSERCOOKIE: 2374
chkChromeAb67Sec: 2324
PugT: 2314
pi: 2156
SyncRTB3: 2086
DPSync3: 1857
IDE: 1584
KRTBCOOKIE_57: 1359
A3: 1345
KRTBCOOKIE_153: 1292
KRTBCOOKIE_377: 1285
i: 1224
ad-id: 1194
```

Total Count of 3P Cookies: 119890

```
-----

Top 15 Third Parties of the 1000 HAR files:
doubleclick: 2730
googleapis: 2378
pubmatic: 2025
rubiconproject: 1903
gstatic: 1389
google: 1245
amazon-adsystem: 1184
yahoo: 960
googlesyndication: 926
googletagmanager: 891
google-analytics: 855
cookielaaw: 823
alicdn: 722
adnxs: 695
adsrvr: 677
```

Total Count of third party domains: 62038